

- 6 Two masses m and $3m$ move towards each other in opposite directions with speeds $2v$ and v respectively.

The masses collide and stick together.

What is the speed of mass m after the collision?

A $\frac{v}{4}$

B $\frac{v}{2}$

C v

D $\frac{5v}{4}$

7

7 A boat floating in fresh water displaces 35.6 kN of water